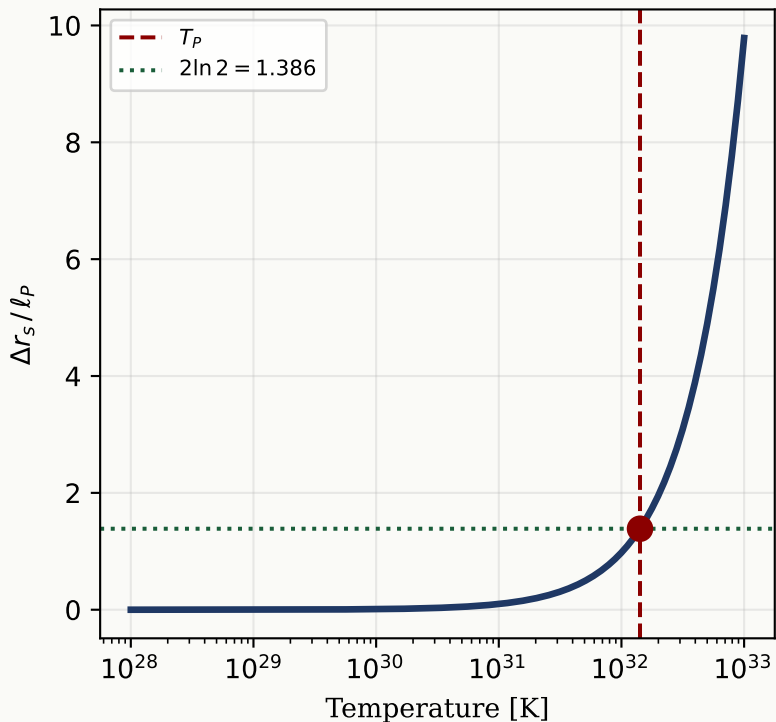


The Planck Coincidence: $\Delta r_s(T_P) = 2 \ln(2) \cdot l_P$ [Exact]

Approach to Planck Coincidence



Algebraic Proof

Planck Coincidence (Exact)

Substitute $T_P = \sqrt{\hbar c^5 / G k_B^2}$:

$$\begin{aligned}\Delta r_s(T_P) &= 2G k_B \ln 2 / c^4 * \sqrt{\hbar c^5 / G k_B^2} \\ &= 2 \ln(2) * \sqrt{\hbar G / c^3} \\ &= 2 \ln(2) * l_P \quad [\text{exact}]\end{aligned}$$

All constants cancel.

Computed: $\Delta r_s(T_P) / l_P = 1.386294$

Expected: $2 \ln(2) = 1.386294$